

CNN-Informer: A hybrid deep learning model for seizure detection on long-term EEG

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ABSTRACT

Timely detecting epileptic seizures can significantly reduce accidental injuries of epilepsy patients and offer a novel intervention approach to improve their quality of life. Investigation on seizure detection based on deep learning models has achieved great success. However, there still remain challenging issues, such as the high computational complexity of the models and overfitting caused by the scarce availability of ictal electroencephalogram (EEG) signals for training. Therefore, we propose a novel end-to-end automatic seizure detection model named CNN-Informer, which leverages the capability of Convolutional Neural Network (CNN) to extract EEG local features of multi-channel EEGs, and the low computational complexity and memory usage ability of the Informer to capture the long-range dependencies. In view of the existence of various artifacts in long-term EEGs, we filter those raw EEGs using Discrete Wavelet Transform (DWT) before feeding them into the proposed CNN-Informer model for feature extraction and classification. Post-processing operations are further employed to achieve the final detection results. Our method is extensively evaluated on the CHB-MIT dataset and SH-SDU dataset with both segment-based and event-based criteria. The experimental outcomes demonstrate the superiority of the proposed CNN-Informer model and its strong generalization ability across two EEG datasets. In addition, the lightweight architecture of CNN-Informer makes it suitable for real-time implementation.

1. Introduction

Epilepsy is a chronic neurological disorder that affects approximately 50 million people worldwide (World Health Organization, 2024). Epileptic seizures are characterized by recurrent and involuntary convulsions, which may be accompanied by loss of consciousness and incontinence (Hussein et al., 2021). These symptoms impose a significant burden on patients and often lead to social stigma and discrimination (Sharmila & Geethanjali, 2016; Yuan et al., 2018). Currently, the diagnosis of epilepsy primarily relies on visually inspecting electroencephalogram (EEG) recordings by neurological specialists (Greene et al., 2008; Zhang et al., 2020). Manually annotating lengthy EEG recordings is a time-consuming and labor-intensive process, increasing the workload of medical staff and challenging to meet the increasing clinical demands. Therefore, the development of automatic seizure detection systems has significant clinical value for alleviating the physical and

economic burdens on patients while significantly reducing the workload of medical experts.

The earliest automatic seizure detection system, proposed by Gotman (1982), employed three parameters derived from the half waves of EEG: slope, rhythmicity, and sharpness, to identify epileptic seizures in clinical settings, laying the vital foundation for automatic seizure detection. To achieve objectives for seizure detection, various signal processing and machine learning techniques have been deployed for automatic seizure detection. Conventional machine learning methods mainly include Support Vector Machine (SVM) (Temko et al., 2011), Random Forest (RF) (Wang et al., 2019), Bayesian classifiers (Yuan et al., 2014) etc. These machine learning methods rely on manual feature extraction, which requires necessary expertise and time-consuming operations. In recent years, deep learning has achieved significant success in computer vision (Hong et al., 2024), image processing (Li et al., 2024a), and natural language processing (Min et al.,

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2024). At the same time, the application of deep neural network models in seizure detection has also become increasingly prevalent, and those models include Convolutional Neural Network (CNN) (Li et al., 2024b), Long Short-Term Memory (LSTM) (Kamakshi & Rengaraj, 2024), and Graph Attention Network (GAT) (Ji et al., 2024). Deep learning methods can automatically extract intrinsic features from input data, enabling end-to-end seizure detection.

In recent years, Transformer has been proposed for natural language processing and computer vision tasks as an alternative to the CNNs. Transformer was introduced by Vaswani et al. (2017) and has been demonstrated tremendous potential in handling time-series due to its ability to capture long-range dependencies in sequential data. The Multi-Head Attention (MHA) mechanism within the Transformer model can process time-series in parallel, avoiding the sequential processing constraints of recursive network structures and significantly improving the efficiency for long time-series data, such as EEG signals. Zhong et al. (2023) employed Stockwell Transform (ST) to attain the time-frequency representations of raw EEG data and utilized Transformer as the classifier to effectively identify epileptic seizures.

However, the self-attention mechanism of the Transformer model is characterized by quadratic time complexity and high memory usage, which increases the deployment costs and limits its applications (Wang et al., 2023b). To overcome these limitations, Zhou et al. (2021) proposed an efficient model for long sequence time-series forecasting (LSTF) based on the improvement of Transformer, named Informer. Informer leverages the ProbSparse self-attention mechanism (Child et al., 2019) to effectively replace traditional self-attention, achieving $\mathcal{O}(\text{LlogL})$ time complexity and memory usage. Moreover, Informer proposes self-attention distilling operation to privilege dominating attention scores in stacking layers, further reducing the total space complexity. The Informer model has been successfully applied in multiple fields such as electric power forecasting (Wang et al., 2023a; Zhou et al., 2021), drought prediction (Shang et al., 2023), medical devices (Sun et al., 2023), aviation flight and air quality prediction (Ma et al., 2023; Wang et al., 2023c), with notable results achieved in these tasks.

EEG signals are non-stationary and susceptible to various noise and artifacts, posing challenges for EEG seizure extraction and accurate seizure detection. Besides, real-time seizure detection is essential in order to issue timely warnings. In this study, we propose a novel end-to-end hybrid model called CNN-Informer for seizure detection, aiming to reduce model complexity and memory usage. Discrete Wavelet Transform (DWT) is employed to filter raw EEGs and these filtered EEGs are fed into CNN-Informer to capture both local and global relations of multi-channel EEGs. The novelty and main contributions of the present work are summarized as follows:

- (1) We propose a novel end-to-end model CNN-Informer for automatic seizure detection, which leverages the capability of CNN to extract local features of multi-channel EEGs, while using Informer to capture the long-range dependencies of these EEGs with low computational complexity and memory usage. To the best of our knowledge, this is the first investigation of combining CNN and Informer for seizure detection.
- (2) The lightweight CNN-Informer utilizes the ProbSparse self-attention mechanism to reduce time complexity and memory usage to the scale of $\mathcal{O}(\text{LlogL})$. Additionally, self-attention distilling operation is employed to extract dominating features, significantly reducing the network size and avoiding the overfitting problem.
- (3) The proposed CNN-Informer model is extensively evaluated on the CHB-MIT dataset and SH-SDU dataset with both segment-based and event-based criteria, with superior results achieved. Our experimental results demonstrate the effectiveness of the proposed CNN-Informer model and its promising potential for real-time seizure detection.

The remainder of this paper is organized as follows. Section 2 presents the related work. Section 3 introduced the two scalp EEG datasets and the methodology for seizure detection. Section 4 delineates the evaluation metrics and experimental results. Section 5 discusses the results and provides the performance comparisons. Finally, this work is concluded in Section 6.

2. Related work

Current techniques for seizure detection can be roughly grouped into two categories: feature engineering methods and deep learning methods (Ke et al., 2022; O'Shea et al., 2020). The following is a review of the most relevant studies.

2.1. Feature engineering methods

Traditional manual feature engineering methods primarily involve the procedure of feature extraction from ictal EEG signals, and these extracted features could serve as the distinctive indicators of seizures. These methods usually deploy time-domain analysis (Klibi et al., 2022; Singh et al., 2023), frequency-domain analysis (Li et al., 2023a, 2021), or nonlinear dynamics analysis (Aaysha et al., 2022) to capture the statistical and dynamical attributes of EEG signals. Due to the non-stationary nature of EEG signals (Guo et al., 2023; Raab et al., 2022), time-frequency analysis approaches were also employed to extract time-frequency features, which include Short Time Fourier Transform (STFT) (Shayeste & Asl, 2023; Yadav et al., 2007), Wavelet Transform (WT) (Chen et al., 2017; Jang & Lee, 2020), Wavelet Packet Decomposition (WPD) (Sairamya et al., 2021; Tafreshi et al., 2006), Empirical Mode Decomposition (EMD) (Moctezuma & Molinas, 2019; Nassir et al., 2023; Tafreshi et al., 2008), and the Stockwell Transform (ST) (Dong et al., 2024a; Geng et al., 2020; Zhang et al., 2018). Additionally, nonlinear dynamical features such as Approximate Entropy (ApEn) (Prasanna et al., 2021) and Fuzzy Entropy (Sukriti & Mitra, 2021) were also leveraged in seizure detection. These extracted features are subsequently fed into a classifier for seizure detection, and commonly utilized classifiers include Support Vector Machines (SVM) (Temko et al., 2011), Random Forests (Wang et al., 2019), Bayesian Classifiers (Yuan et al., 2014), and so on. However, the non-stationary nature of EEG signals and the existing noise and artifacts pose difficulties for hand-crafted feature extraction and machine learning based seizure detection.

2.2. Deep learning methods

With the rapid advancement of deep learning technology, deep neural networks have been increasingly adopted for seizure detection tasks. For instance, Yan et al. (2019) proposed an automated seizure detection algorithm based on CNN that utilized the EEG spectrum as the feature to effectively distinguish seizure and non-seizure states. Similarly, Raghu et al. (2020) converted raw EEG data into spectrograms, and fed them into a CNN network trained with transfer learning, realizing classification of epileptic seizure types. Furthermore, Liu et al. (2024) presented a novel Cosine Convolutional Neural Network (CosCNN), which replaces traditional kernels with cosine kernels modulated by only two learnable factors, effectively addressing the common issues of high memory consumption, kernel redundancy, and overfitting in traditional CNNs, and achieving excellent performance in seizure detection. To better capture the spatial and temporal dependency of seizure EEG activity, Li et al. (2022) proposed a spatio-temporal-spectral hierarchical graph convolutional network with an active preictal interval learning scheme, which constructed hierarchical graphs of seizure EEG rhythms to represent temporal dependency and spatial coupling and achieve accurate seizure detection. In addition, Hu et al. (2020) employed Bidirectional Long Short-Term Memory (Bi-LSTM) to capture the temporal dependency of the EEG time series,

realizing automatic seizure detection. Song et al. (2024) combined Residual Convolution (ResCon) and LSTM to address the issue of gradient explosion and explore inherent seizure features, realizing cross-patient seizure detection. To prevent the model overfitting, Liu et al. (2022) proposed to randomly shuffle EEG channels by channel-perturbation, successfully performing the cross-patient seizure detection using CNN and Bi-LSTM.

Due to the ability to capture long-range dependencies in sequential data while processing long time-series data in parallel, the Transformer and Multi-Head Attention (MHA) have been applied in seizure detection. Zhu and Wang (2023) developed an epileptic seizure detection method for multi-channel EEG signals based on Transformer. Liu et al. (2023) developed MHA augmented CNN algorithm that effectively addresses the limitations of CNN in capturing global information of input signals, achieving proficient early seizure prediction. Sun et al. (2022) combined CNN and Transformer to capture long-term continuous EEG features, achieving end-to-end seizure detection. The performance of deep learning networks heavily relies on large model scale and expensive training, which may lead to model redundancy and overfitting, especially when the training dataset is limited.

3. Material and methods

In this section, we describe the datasets and methods employed in this study. Firstly, we introduce the CHB-MIT dataset and the SH-SDU dataset. Then, we provide the framework of the proposed seizure detection method, including: pre-processing, the CNN-Informer model, and post-processing.

3.1. Scalp EEG dataset

The publicly available CHB-MIT dataset was collected at the Children's Hospital Boston (CHB) and the Massachusetts Institute of Technology (MIT), encompassing 24 cases of EEG recordings from 23 pediatric patients with intractable epileptic seizures (Nassir et al., 2023). All signals have a sampling rate of 256 Hz and a resolution of 16-bit. The international 10–20 System for electrode positioning in bipolar configuration was used for these recordings (Prasanna et al., 2021). The dataset contains different seizure types, such as focal, lateral and generalized seizures.

The CHB-MIT dataset contains approximately 983 h of EEG recordings, covering 198 epileptic seizure events with a total duration of about 11,536 s. The onset and offset of each seizure were annotated by professional clinical physicians. Details of the CHB-MIT EEG dataset are presented in Table 1. In this study, to ensure the consistency of the input feature maps, we selected 18 common EEG channels from the recordings of each patient: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, and CZ-PZ. Our proposed model was evaluated on EEG recordings from all patients except Chb6 and Chb16, due to their seizure events being mostly shorter than 15 s.

Another EEG dataset was collected at the Second Hospital of Shandong University (SH-SDU) in Jinan, China, originating from 10 adult patients diagnosed with epilepsy. The EEG recordings were initially sampled at a rate of 500 Hz, which were down-sampled to 256 Hz in this study. Approximately 143 h of EEG recordings were annotated, including 103 epileptic seizure events with a total duration of about 9898 s. The onset and offset of each seizure were marked by professional clinical physicians in seconds. Similarly, 19 common EEG channels were selected for each patient, including Fp1, Fp2, F3, F4, C3, C4, P3, P4, Cz, Pz, Fz, O1, O2, F7, F8, T3, T4, T5, and T6. The description of the SH-SDU EEG dataset is shown in Table 2.

Table 1

Details of the used CHB-MIT EEG dataset.

Case	Gender	Age	Number of seizure events	Duration of epileptic seizures (s)
Chb01	F	11	7	442
Chb02	M	11	3	172
Chb03	F	14	7	402
Chb04	M	22	4	378
Chb05	F	7	5	558
Chb06	F	1.5	10	138
Chb07	F	14.5	3	325
Chb08	M	3.5	5	919
Chb09	F	10	4	276
Chb10	M	3	7	447
Chb11	F	12	3	806
Chb12	F	2	40	1475
Chb13	F	3	12	535
Chb14	F	9	8	109
Chb15	M	16	20	1992
Chb16	F	7	10	84
Chb17	F	12	3	293
Chb18	F	18	6	317
Chb19	F	19	3	236
Chb20	F	6	8	294
Chb21	F	13	4	199
Chb22	F	9	3	204
Chb23	F	6	7	424
Chb24	—	—	16	511
Total	—	—	198	11,536

Table 2

Details of the used SH-SDU EEG dataset.

Case	Gender	Age	Number of seizure events	Duration of epileptic seizures (s)
Pat01	F	28	16	601
Pat02	M	61	10	2208
Pat03	M	72	6	532
Pat04	F	37	6	138
Pat05	M	49	17	1876
Pat06	F	69	7	868
Pat07	M	78	4	375
Pat08	F	58	10	742
Pat09	M	71	3	895
Pat10	M	64	24	1663
Total	—	—	103	9898

3.2. Methodology

The framework of the proposed automated seizure detection system is illustrated in Fig. 1, which encompasses three parts: Pre-processing, Model construction, and Post-processing. In the pre-processing phase, segmentation and DWT are employed to enhance signal quality. The model is constructed by integrating CNN with Informer. Post-processing involving a moving average filter (MAF), thresholding, and the collar technique is applied to further improve the detection performance. Detailed explanations of each part are provided in the subsequent sections.

3.2.1. Pre-processing

In this work, the long-term EEG recordings are analyzed by a moving-window technique. The EEG data are separated into 4 s epochs by a sliding window. Given that the duration of seizure recordings is typically much shorter than that of non-seizure periods, seizure data and non-seizure data are significantly imbalanced in seizure detection tasks. To balance the data and augment the data scale of seizure EEG recordings, we employ a 4 s sliding window method with 50% overlap for segmenting the seizure EEG recordings. For non-seizure EEG recordings, we use non-overlapping 4 s sliding windows for segmentation. After pre-

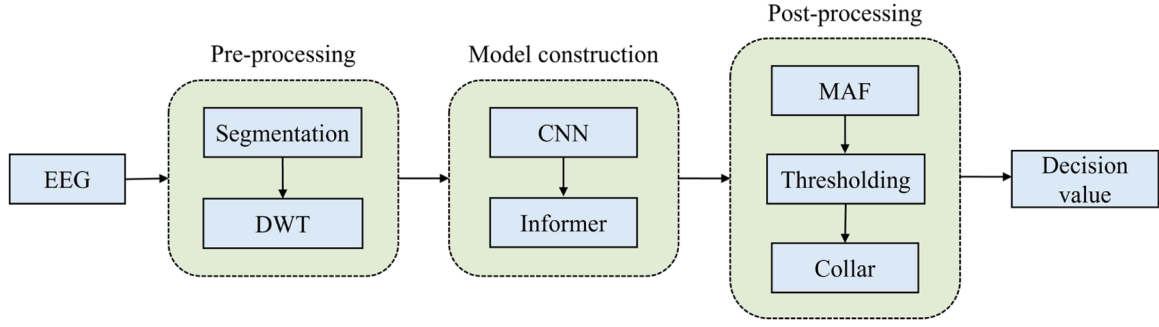


Fig. 1. The framework of the proposed method for seizure detection.

processing, each EEG segment has a size of (18, 1024), where 18 represents the number of channels, and 1024 corresponds to the number of sampling points.

In order to further improve the quality of EEG signals, we need to filter the signals to remove noise and artifacts. As an excellent tool for time-frequency analysis, wavelet transform has been widely adopted in the realm of signal processing. Wavelet analysis deploys variable time-frequency windows to provide an optimal time-frequency resolution, offering an effective solution to the limitations of the Fourier transform. In this work, the DWT is employed to effectively filter out ocular artifacts and power line noise from EEG signals, thereby enhancing the quality of EEG signals related to seizures. Due to the resemblance in shape between the Daubechies-4 (Db4) wavelet and the characteristic wave of seizure EEG (Khan & Gotman, 2003; Sharmila & Geethanjali, 2016; Yuan et al., 2018), the Db4 wavelet is selected as the mother wavelet in this work. Each EEG segment with a sampling frequency of 256 Hz is decomposed into 5 scales using DWT, corresponding to the following frequency ranges: 64–128 Hz (cD1), 32–64 Hz (cD2), 16–32 Hz (cD3), 8–16 Hz (cD4), 4–8 Hz (cD5), and 0–4 Hz (cA5). Previous studies found that the EEG frequency of epileptic seizures mainly focuses on 3–29 Hz (Aziz et al., 1986; Gotman, 1982). Therefore, cD5, cD4, cD3 and cA5 are reconstructed to obtain the filtered EEG. Fig. 2 illustrates the process of EEG signal decomposition and reconstruction using DWT.

3.2.2. CNN-Informer model

The proposed CNN-Informer incorporates CNN module with Informer encoder to achieve end-to-end seizure detection. The architecture of the CNN-Informer model is shown in Fig. 3, and its algorithm is illustrated in Algorithm 1. Based on the local convolution operator, CNN has demonstrated its powerful capability in automatic feature extraction from images (Ilesanmi & Ilesanmi, 2021; Zhou et al., 2023a). In the CNN-Informer model, we deploy a CNN module to extract local features from multi-channel EEGs processed by DWT to obtain the compressed feature matrix. The CNN module is presented in Fig. 4, which includes three layers of one-dimensional convolution. The first two layers have 18 filters of size 4 with a stride of 2, and the third layer has 64 filters of size 3 with padding of 1. The first two layers focus on feature extraction, while the third layer acts as a channel separator for more detailed feature detection.

To accelerate network convergence, batch normalization (BN) is applied after the convolutional layers. We use the ELU activation function to bring the gradients closer to natural gradients and accelerate the learning process. The dropout layer is included for robustness, and a max-pooling layer of size 2 is used after each of the first two convolutional layers to reduce feature map dimensions. The model is trained by the Adam optimizer with a learning rate of 10^{-4} and a batch size of 32. Finally, the CNN outputs a feature map with a size of (64,63), which will be sent to the Informer for further processing.

Informer is a deep learning architecture based on the Transformer model, specifically designed for LSTF. Compared to the traditional RNN-based model, the Transformer relies on the self-attention mechanism

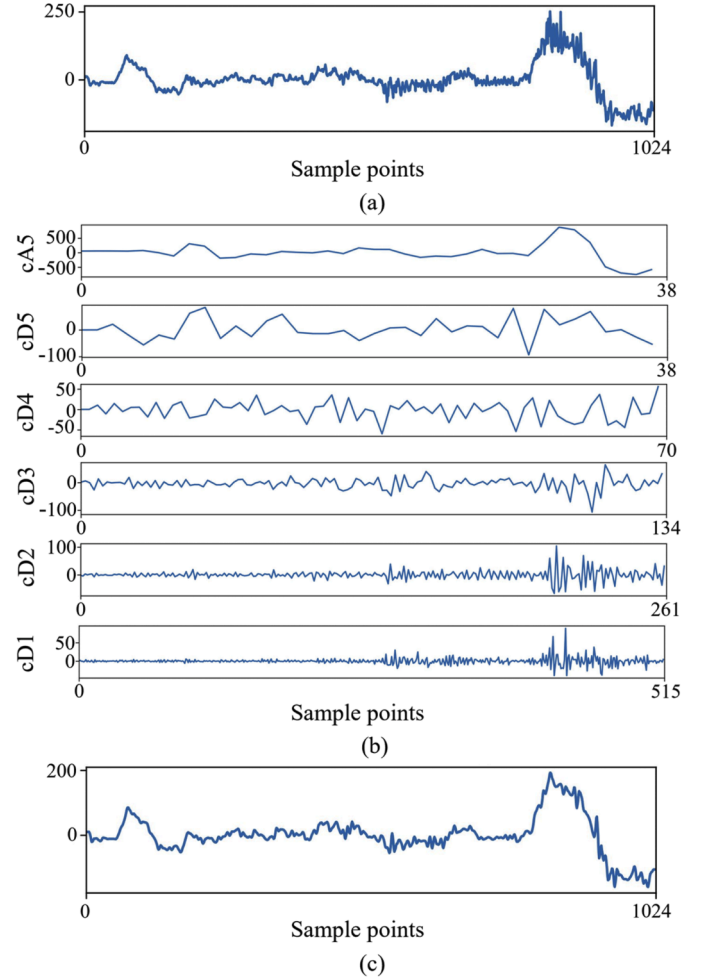


Fig. 2. Filtering EEG signal using DWT. (a) Original 4 s EEG signal. (b) EEG decomposition using DWT. (c) EEG reconstruction using DWT.

which is beneficial to process data in parallel and capture global dependency, effectively addressing the issue of information loss in long sequence processing. However, the self-attention mechanism in the Transformer model has time complexity and memory usage of $\mathcal{O}(L^2)$, which requires a high computational cost. To address this issue, Zhou et al. (2021) proposed an efficient model called Informer.

One of the core features of the Informer is the ProbSparse self-attention mechanism, which was introduced based on the observation that the distribution of self-attention probability exists potential sparsity. This distribution is typically long-tailed, meaning that during the computation of attention, only a few query-key dot-product pairs (dominant dot-product pairs) contribute significantly to the final

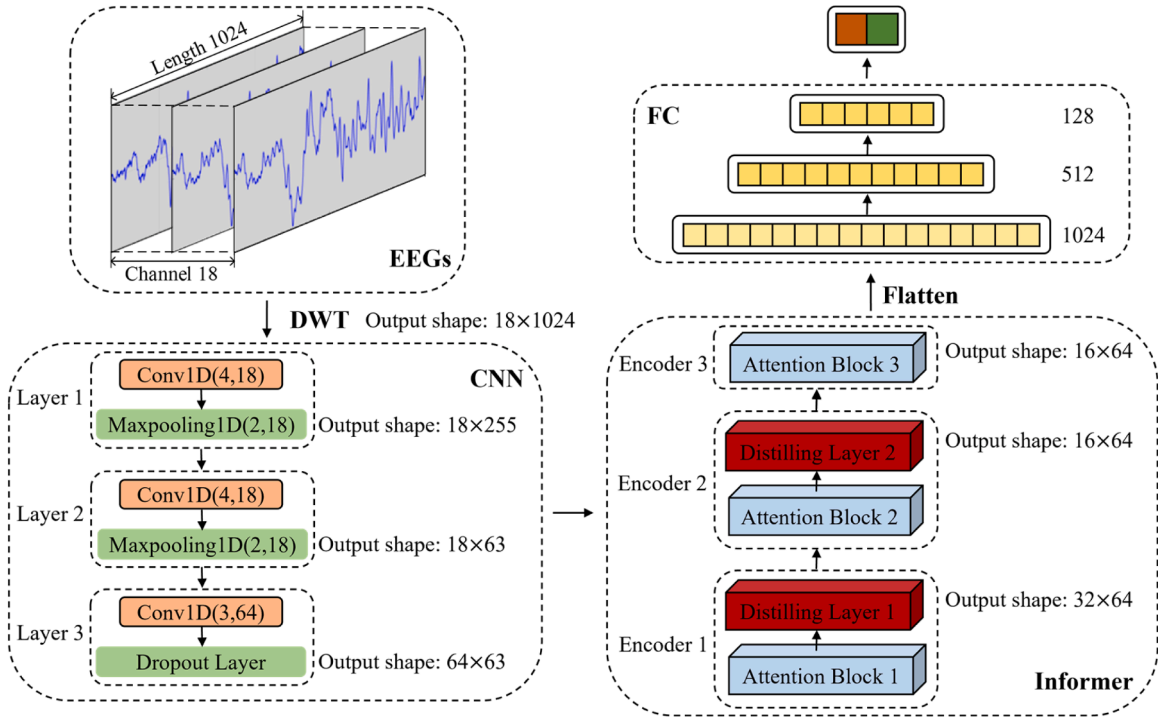


Fig. 3. Proposed CNN-Informer architecture and flow diagram.

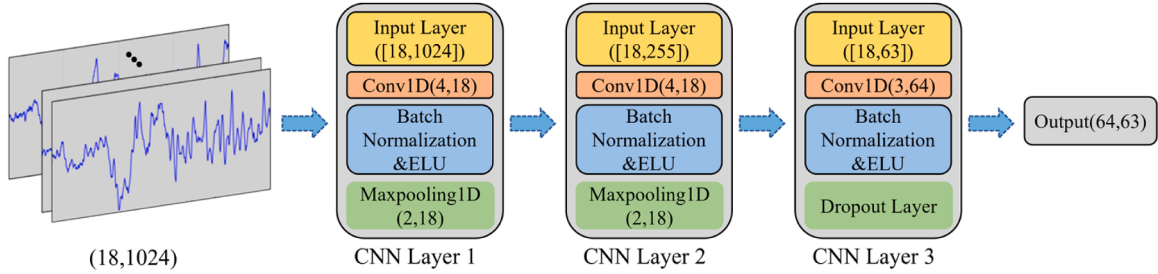


Fig. 4. Structure of the CNN module.

attention scores, while the majority of other dot-product pairs (inferior dot-product pairs) contribute very little to the attention. Informer uses a sparsity measure of the queries to select and retain those most important queries, significantly reducing the computational load, and bringing the time complexity and memory usage of the self-attention mechanism down to $\mathcal{O}(L \log L)$.

Let Q , K and V represent the query matrix, the key matrix and the value matrix in the self-attention mechanism, here $Q \in \mathbb{R}^{L_q \times d_q}$, $K \in \mathbb{R}^{L_k \times d_k}$, $V \in \mathbb{R}^{L_v \times d_v}$. In the Informer, the sparsity measurement of the i -th query is defined as:

$$\bar{M}(q_i, K) = \max_j \left\{ \frac{q_i k_j^T}{\sqrt{d}} \right\} - \frac{1}{L_k} \sum_{j=1}^{L_k} \frac{q_i k_j^T}{\sqrt{d}}, \quad (1)$$

where the terms q_i and k_i denote the i -th row in Q , K , correspondingly. d represents the dimension of the input sequence, L is the length of the input sequence. The ProbSparse self-attention mechanism is realized by allowing each key to only focus on the u dominant queries:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{\bar{Q} K^T}{\sqrt{d_k}} \right) V, \quad (2)$$

where $\bar{Q}$ is the matrix composed of the top- u q_i , with $\bar{Q} \in \mathbb{R}^{u \times d_q}$. u can be obtained with:

$$u = c \times (\ln L_q), \quad (3)$$

here u is controlled by a constant sampling factor c .

For the q_i that are not selected, the attention scores at their corresponding positions are replaced with the average value, ensuring that the length of the input and output sequences remains unchanged. The ProbSparse self-attention procedure is illustrated in Algorithm 2.

Since the characteristics of EEG signals are basically non-stationary and highly stochastic, there is no need to introduce additional time-stamps, and token embedding alone is sufficient (Sun et al., 2023). Furthermore, the last convolution layer in our CNN-Informer model is used to separate the input channels and serves as the embedding layer. In this study, only the encoder module is employed for the seizure detection task. The overview of the Informer module is shown in Fig. 5.

Additionally, a self-attention distilling operation is introduced to eliminate the redundancy in the encoder's feature map in the Informer. This operation employs a combination of convolutional and pooling layers to halve the length of the input sequence from the previous layer. It aims to extract the dominating features while significantly reducing the memory size and addressing the memory bottleneck in stacking layers for long inputs, thereby mitigating the overfitting. The self-attention distilling operation is as follows:

Algorithm 1

CNN-Informer.

Input: $X \in \mathbb{R}^{C \times L}$
Output: Predicted label

1. **For** Conv_layer = 1 to 3 **do**
2. Set kernel_size, stride, and padding for Conv_layer
3. Compute the one-dimensional convolution to obtain $\hat{F}_{Conv} \in \mathbb{R}^{\hat{C} \times L}$
4. Apply the activation function (ELU) to $\hat{F}_{Conv}$
5. **If** Conv_layer < 3 **then**
6. Compute the one-dimensional max pooling to obtain $\tilde{F}_{Conv} \in \mathbb{R}^{\tilde{C} \times L}$
7. **End If**
8. **End for**
9. **For** Enc_layer = 1 to 3 **do**
10. Compute multi-head ProbSparse self-attention
11. **If** Enc_layer < 3 **then**
12. Compute self-attention distilling operation
13. **End If**
14. Set residual connection and layer normalization
15. Compute the feed-forward neural network
16. **End for**
17. Compute the fully connected layer
18. Compute the softmax mapping
19. Find the maximum value and its corresponding label
20. **Return** Predicted label

Algorithm 2

ProbSparse self-attention

Input: $Q \in \mathbb{R}^{m \times d}, K \in \mathbb{R}^{n \times d}, V \in \mathbb{R}^{n \times d}$
Output: self-attention feature map S

1. Set constant sampling factor $c, u = c \ln m, U = m \ln n$
2. randomly select U dot-product pairs from K as $\bar{K}$
3. Compute the sample score $\bar{S} = Q\bar{K}^T$
4. Compute the measurement $M = \max(\bar{S}) - \text{mean}(\bar{S})$ by row
5. Select top- u queries under M to compose $\bar{Q}$
6. Compute $S_1 = \text{softmax}(\bar{Q}\bar{K}^T / \sqrt{d})V$
7. Compute $S_0 = \text{mean}(V)$
8. Compute $S = \{S_1, S_0\}$ by their original rows accordingly
9. **Return** S

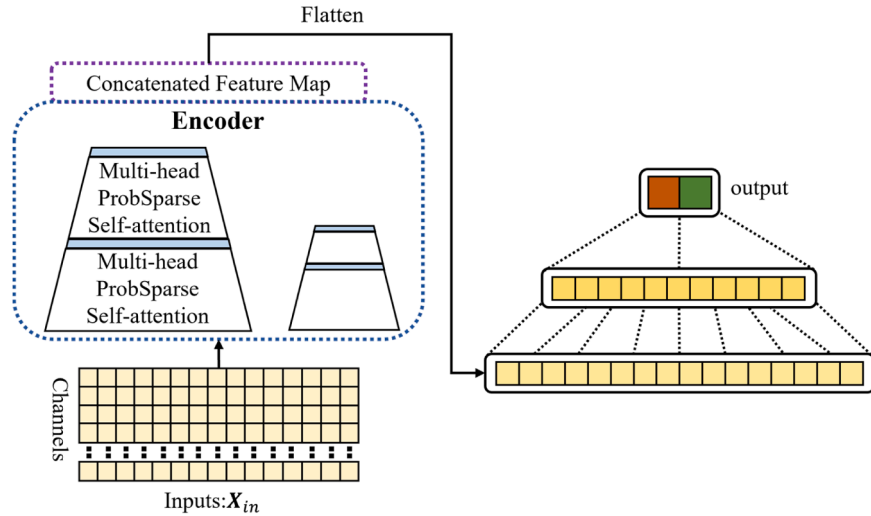


Fig. 5. Overview of the Informer module. The blue trapezoid represents the self-attention distilling operation.

$$X_{j+1}^t = \text{Maxpool}\left(\text{ELU}\left(\text{Conv1d}\left(\left[X_j^t\right]_{AB}\right)\right)\right), \quad (4)$$

where X_{j+1}^t is the output of the ProbSparse self-attention mechanism at that layer, and $\left[X_j^t\right]_{AB}$ denotes the output of the previous layer. As illustrated in Fig. 6, the Informer encoder module used in this study employs three ProbSparse multi-head self-attention blocks, with self-

attention distilling operations embedded between the layers.

3.2.3. Post-processing

In order to reduce isolated false positives or negatives, post-processing is implemented on the output of the CNN-Informer model. This post-processing involves moving average filter (MAF), thresholding, and collar technique. The MAF is used to process the model's

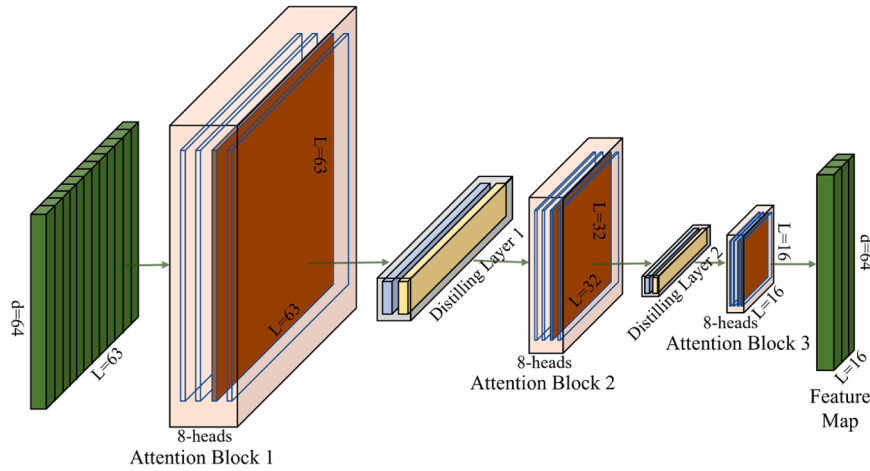


Fig. 6. Structure of the Informer encoder.

output scores, aiming to mitigate the impact of random noise and sharp fluctuations. The filter is defined as follows:

$$\mathcal{Y}_t = \frac{1}{2N+1} \sum_{n=-N}^N s_{t+n}, \quad (5)$$

where $\mathcal{Y}_t$ denotes the output score after MAF, $2N+1$ represents the length of the MAF, and s represents the output score from the CNN-Informer network. The scores are then compared with a threshold value to obtain binary labels: 1 for epileptic seizure segments and 0 for normal segments. Although MAF can reduce the false detection rate (FDR), it will cause a delay in the seizure onset detection. Therefore, it is necessary to compensate for the binary scores using collar technique. By employing the collar technique, the detected seizure segment is extended by K points at both its beginning and end. Due to the significant specificity of EEG signal for each patient, the value of K is adjusted for each patient to achieve optimal results. Fig. 7 illustrates the post-processing process.

4. Result

All experiments were conducted on an Ubuntu system configured with a 12th Gen Intel(R) Core(TM) i9-12900 K processor with 32GB of RAM operating at frequency of 3.20G Hz, and GeForce RTX 3080 GPU. The code was developed in the PyCharm utilizing the PyTorch framework and the Python programming language.

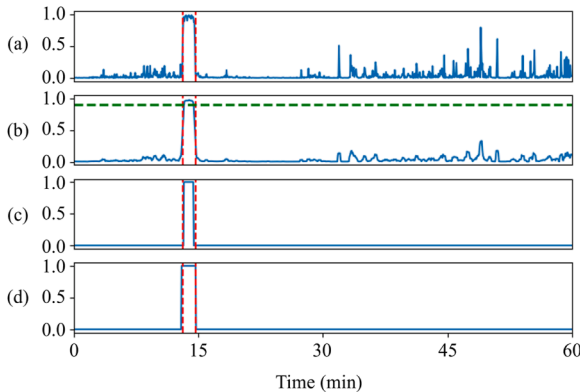


Fig. 7. The post-processing procedure. The area between the two red vertical lines corresponds to the epileptic seizure signals annotated by experts, and the green horizontal line represents the selected threshold. (a) The output scores of classification network. (b) The output of MAF. (c) The binary decision after thresholding. (d) The final labels after applying the collar technique.

We adopt two types of assessment criteria for model performance evaluation, namely the segment-based and event-based criteria. For the segment-based evaluation criteria, three performance metrics are used: sensitivity, specificity, and accuracy. The corresponding formulas for these metrics are as follows:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100\%, \quad (6)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100\%, \quad (7)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%, \quad (8)$$

where TP (True Positive) refers to the number of epileptic seizure segments correctly detected by the system, TN (True Negative) is the number of non-seizure segments correctly identified by the system, FP (False Positive) denotes the number of non-seizure segments incorrectly classified as seizure segments, and FN (False Negative) indicates the number of seizure segments are incorrectly classified as non-seizure segments. The event-based evaluation metrics are mainly used to measure the feasibility of the system in clinical applications (Liu et al., 2020). In the event-based evaluation criteria, three performance metrics are utilized: sensitivity, false detection rate (FDR), and latency. Sensitivity refers to the percentage of correctly detected epileptic seizure events out of the total number of seizure events. FDR represents the number of seizure events falsely detected per hour, and latency is the interval between the onset time of a seizure as marked by experts and the onset time detected by the model.

The segment-based evaluation results on the CHB-MIT dataset are shown in Table 3, where the sensitivity reaches 100% in up to 20 cases, with an average sensitivity of 99.54%. The specificity higher than 99% are achieved on 14 patients, with the highest specificity reaching 99.98%, and the average specificity is 98.55%. The proposed method also achieves a high average accuracy of 98.54% for 22 patients, among which 14 patients have an accuracy of over 99%, and only one patient has an accuracy below 95%. On the other hand, the event-based evaluation results on the CHB-MIT dataset are shown in Table 4. The sensitivity of 22 patients reached 100%, and only 2 patients have seizure events that were not fully detected. Out of 109 seizure events, 107 were correctly detected, with an average sensitivity of 99.07%. Additionally, the average FDR is 0.16/h, and the average latency is 22.21s. The testing time for one hour of EEG is 1.76 s.

The segment-based evaluation results on the SH-SDU dataset are presented in Table 5. The sensitivity of 100% is observed in 7 cases, and the average sensitivity was 95.40%. For most of the cases, specificity and

Table 3

The results of segment-based evaluation on CHB-MIT dataset.

Case	Sensitivity (%)	Specificity (%)	Accuracy (%)
Chb01	100.00	99.58	99.58
Chb02	100.00	99.98	99.98
Chb03	100.00	98.97	98.97
Chb04	100.00	99.28	99.28
Chb05	100.00	99.77	99.77
Chb07	100.00	93.92	93.60
Chb08	100.00	99.29	99.30
Chb09	100.00	99.94	99.94
Chb10	100.00	99.46	99.47
Chb11	100.00	99.97	99.97
Chb12	95.55	95.02	95.02
Chb13	94.29	99.27	99.25
Chb14	100.00	99.62	99.62
Chb15	100.00	95.23	95.29
Chb17	100.00	98.64	98.65
Chb18	100.00	99.72	99.72
Chb19	100.00	99.89	99.89
Chb20	100.00	97.31	97.31
Chb21	100.00	99.19	99.19
Chb22	100.00	99.93	99.93
Chb23	100.00	98.64	98.64
Chb24	100.00	95.38	95.45
Average	99.54	98.55	98.54

accuracy exceeded 90%, resulting in average specificity of 92.92% and accuracy of 92.78%. Table 6 presents the event-based evaluation results on the SH-SDU dataset, where sensitivity reached 100% in 7 cases, while only 3 cases report missed detection. Among the 73 epileptic events, 66 are correctly detected with average sensitivity of 94.83%, FDR of 1.24/h and latency of 33 .81s.

5. Discussion

In this section, we analyze and compare the proposed CNN-Informer model from multiple aspects to validate the effectiveness and superiority of the proposed seizure detection method. First, we perform feature visualization to intuitively analyze the impact of the ProbSparse self-attention and distilling operation. Then, we conduct two types of ablation studies to analyze the influence of model configuration and parameters. Additionally, we analyze the complexity of the model. Finally, we compare the proposed method with other state-of-the-art methods.

Table 4

The results of event-based evaluation on CHB-MIT dataset.

Case	Duration of test set (h)	Number of testing seizures	Number of true detections	Sensitivity (%)	FDR (/h)	Latency (s)
Chb01	39.91	6	6	100.00	0.13	17.83
Chb02	33.27	1	1	100.00	0.00	38.00
Chb03	35.00	4	4	100.00	0.17	15.25
Chb04	122.23	2	2	100.00	0.09	67.50
Chb05	38.00	4	4	100.00	0.00	33.25
Chb07	63.05	2	2	100.00	0.03	15.50
Chb08	19.01	4	4	100.00	0.00	34.00
Chb09	66.28	3	3	100.00	0.00	18.00
Chb10	44.02	4	4	100.00	0.07	23.00
Chb11	34.00	2	2	100.00	0.00	5.50
Chb12	17.01	16	15	93.75	0.82	8.25
Chb13	30.00	7	6	85.71	0.17	15.80
Chb14	22.00	4	4	100.00	0.09	8.50
Chb15	37.01	17	17	100.00	0.32	23.18
Chb17	20.01	2	2	100.00	0.25	18.50
Chb18	33.63	4	4	100.00	0.00	27.25
Chb19	29.00	2	2	100.00	0.00	26.50
Chb20	25.60	5	5	100.00	0.74	19.00
Chb21	30.83	2	2	100.00	0.10	8.50
Chb22	29.00	1	1	100.00	0.00	33.00
Chb23	24.48	6	6	100.00	0.29	20.67
Chb24	7.00	11	11	100.00	0.14	11.55
Total	800.34	109	107	99.07	0.16	22.21

5.1. Feature visualization

In this work, we propose a novel model named CNN-Informer for automatic seizure detection. Firstly, the CNN module is used to extract features and separate channels from the pre-processed signals, then the feature matrix is fed into the Informer encoder. The core of the Informer encoder lies in the ProbSparse self-attention mechanism, which efficiently extracts dominant attention, while significantly reducing memory usage and computational complexity. Moreover, the inclusion of the self-attention distilling operation further diminishes the model memory

Table 5

The results of segment-based evaluation on SH-SDU dataset.

Case	Sensitivity (%)	Specificity (%)	Accuracy (%)
Pat01	89.39	86.62	86.09
Pat02	100.00	95.17	95.36
Pat03	100.00	97.04	97.05
Pat04	100.00	83.55	83.67
Pat05	100.00	93.56	94.03
Pat06	100.00	99.10	98.94
Pat07	100.00	97.01	97.01
Pat08	88.59	86.26	86.40
Pat09	100.00	98.02	98.04
Pat10	76.00	92.84	91.20
Average	95.40	92.92	92.78

Table 6

The results of event-based evaluation on SH-SDU dataset.

Case	Duration of test set (h)	Number of testing seizures	Number of true detections	Sensitivity (%)	FDR (/h)	Latency (s)
Pat01	17.07	10	9	90.00	1.33	21.88
Pat02	16.04	8	8	100.00	0.17	50.50
Pat03	15.56	3	3	100.00	0.74	45.33
Pat04	3.77	3	3	100.00	2.77	29.00
Pat05	12	14	14	100.00	2.17	34.00
Pat06	3.25	5	5	100.00	1.27	31.20
Pat07	18.18	2	2	100.00	0.61	2.00
Pat08	16.04	6	5	83.33	1.36	35.00
Pat09	17.51	2	2	100.00	0.17	33.75
Pat10	23.99	20	15	75.00	1.82	55.47
Total	143.41	73	66	94.83	1.24	33.81

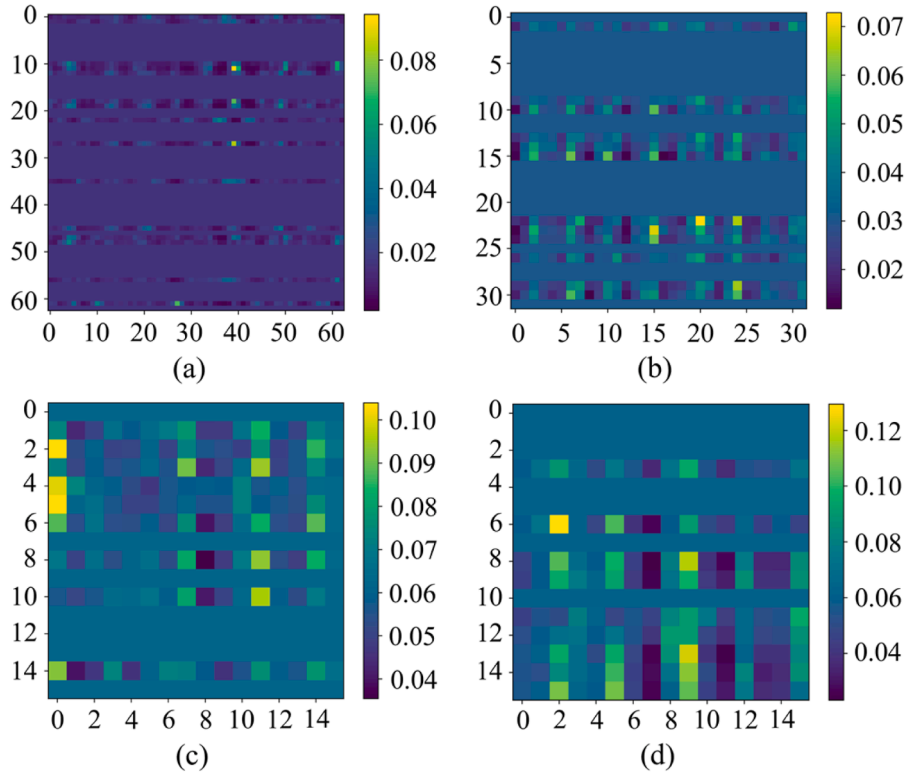


Fig. 8. Visualization of the self-attention matrix. (a) The self-attention matrix after passing through the first ProbSparse self-attention block. (b) The self-attention matrix after passing through the first self-attention distilling operation. (c) The self-attention matrix after passing through the second ProbSparse self-attention block and the second self-attention distilling operation. (d) The self-attention matrix after passing through the third ProbSparse self-attention block.

usage. The visualization of the self-attention matrix is shown in Fig. 8. As depicted in Fig. 8, after extracting dominant attention, the inferior attention is replaced with the average value. Furthermore, the length of the feature matrix is halved after the attention distilling operation. It is found from Fig. 8(a), (b), and (c) that after applying ProbSparse self-attention and distilling operation, the proportion of dominant attention in the attention matrix increases. By comparing Fig. 8(c) and (d), the dominant attention is further enhanced through the third ProbSparse self-attention block.

5.2. Ablation study

Informer is a lightweight network derived from Transformer. To further demonstrate the CNN-Informer model's superiority, we conducted two sets of ablation experiments. Additionally, we performed an ablation analysis on the selection of the sampling factor c in the Informer.

5.2.1. Model analysis

The performance comparison with different models on the CHB-MIT dataset is shown in Table 7. M-01 is a model that uses CNN and a standard Transformer for seizure detection. M-02 combines CNN with Informer without the self-attention distilling operation. M-03 represents the proposed CNN-Informer model. The performance evaluation

outcomes, including segment-based metrics and event-based metrics in Table 7, show that the CNN-Informer model (M-03) achieves higher accuracy, specificity, and sensitivity, as well as shorter detection latency. The FDR of the CNN-Informer model is 0.16, which is significantly lower than M-01 and M-02. Those indicate that the CNN-Informer model demonstrates its superior seizure detection capability.

To intuitively assess the classification capability of the three models in feature extraction, we generated t-Distributed Stochastic Neighbor Embedding (t-SNE) maps of the EEG features extracted by each model from the raw EEG recordings, as shown in Fig. 9. Concretely, Fig. 9(a) and (b) depicted the t-SNE maps of the ictal and interictal EEG samples and their features extracted by CNN. The t-SNE maps of the features extracted by the M-01 model, M-02 model, and M-03 model are illustrated in Fig. 9(c)–(e), respectively. As shown in Fig. 9(b), the features extracted by CNN yield poor distinguishability, unable to meet the ictal EEG classification requirement. Comparing (c), (d), and (e) in Fig. 9, it is found that although the M-01 model attains good separability on the two types of EEG samples, the boundaries between them are unclear, resulting in a lower level of distinguishability compared to the M-03 model. The M-02 model has relatively better classification boundaries, but fails to concentrate similar samples. In contrast, the M-03 model provides the best separable performance, revealing its effectiveness of the proposed CNN-Informer model for seizure detection.

At the same time, to further illustrate the relationship between

Table 7
Comparison of three groups of models on CHB-MIT dataset.

Model	Segment-based			Event-based		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Sensitivity (%)	FDR (/h)	Latency (s)
M-01	97.18	98.38	97.32	97.96	0.71	22.87
M-02	97.51	96.38	97.57	96.63	1.20	22.50
M-03	98.54	99.54	98.55	99.07	0.16	22.21

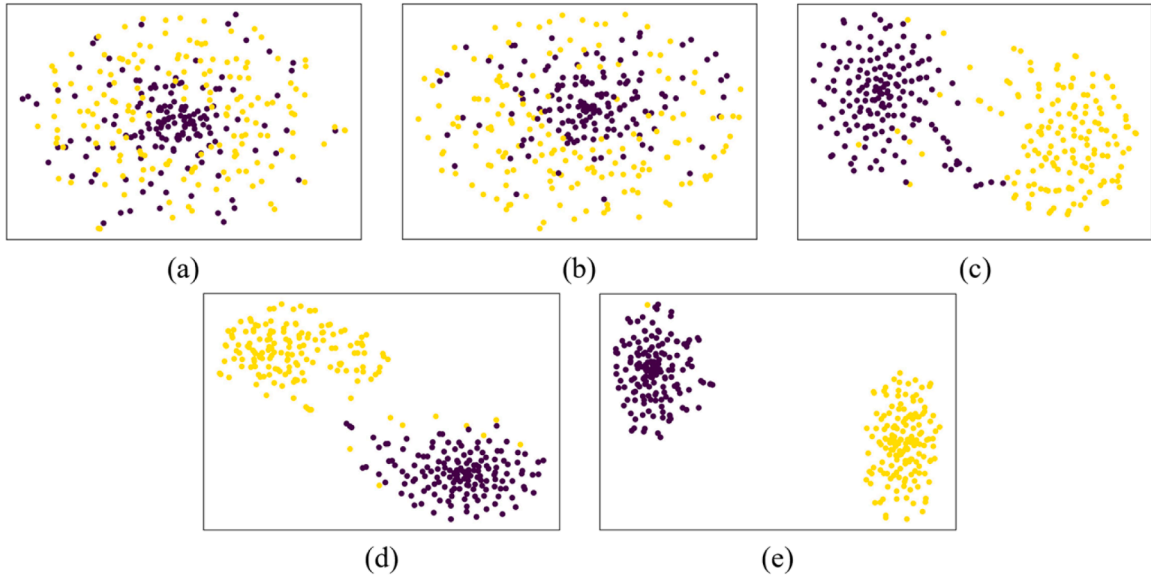


Fig. 9. t-SNE maps of features extracted by CNN, Transformer and Informer modules. (a) The raw two-class EEG samples, with yellow dots indicating epileptic seizure samples and purple dots indicating normal samples. (b) The features extracted by the CNN. (c) The features extracted by the CNN-Transformer (M-01). (d) The features extracted by the CNN-Informer without the self-attention distilling operation (M-02). (e) The features extracted by the proposed CNN-Informer (M-03).

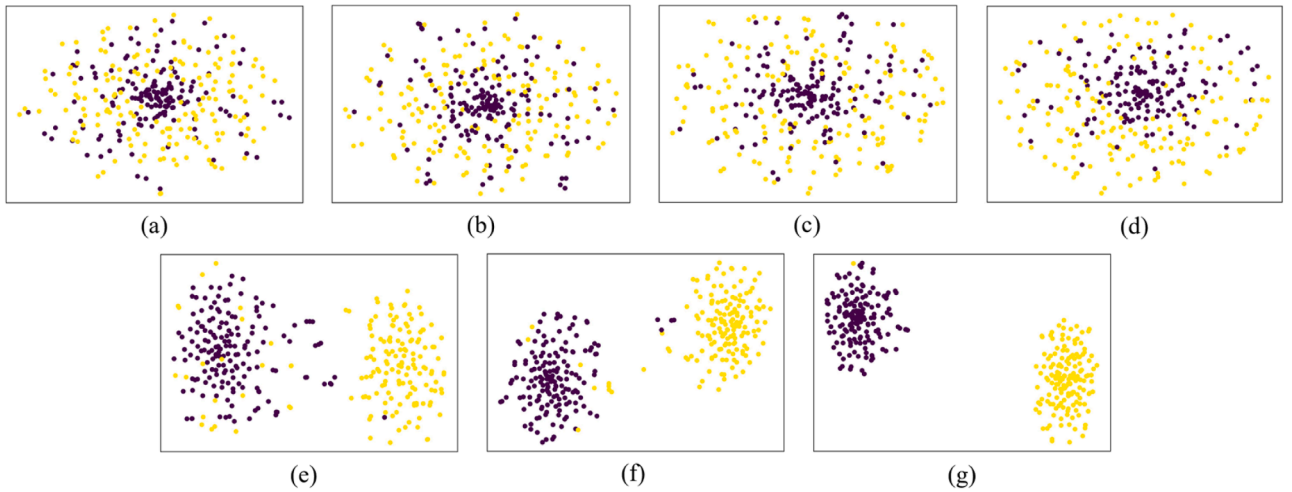


Fig. 10. t-SNE maps of features extracted by each layer of the proposed CNN-Informer model. (a) The raw two-class EEG samples, with yellow dots indicating epileptic seizure samples and purple dots indicating normal samples. (b), (c), and (d) correspond to the features extracted after three CNN layers, respectively. (e), (f), and (g) represent the features extracted after three Informer Encoder layers, respectively.

seizure signal feature maps and the model size, we employed t-SNE plots to visualize the clustering of the EEG samples after each layer of the proposed CNN-Informer model, as shown in Fig. 10. Fig. 10(a) presents the distribution of two-class raw EEG samples. Fig. 10(b), (c), and (d) depict the feature maps after three CNN layers, respectively, and the CNN module serves as a front-end to capture the local features and reduce their dimensionality. Fig. 10(e), (f), and (g) show the feature maps after three Informer Encoder layers, respectively. It is found that the features of ictal and non-ictal samples are gradually separated into two clusters after three Informer Encoder layers, demonstrating their ability to conduct the multi-channel EEG classification during epileptic seizures with high accuracy.

5.2.2. Parameter analysis

Furthermore, in order to assess the impact of the constant sampling factor c in Eq. (3), we conduct ablation experiments with c values set at 1, 3, 5, 7, and 15 for the same input data length. As illustrated in Fig. 11,

the results demonstrate that the model performs best at $c=3$ for the first four evaluation metrics. For different c values, FDR exhibits greater variability, and the model attains a lower false detection rate when $c=3$. This is primarily because choosing a smaller c value leads to some loss of EEG information, whereas a larger c value causes the model to focus on redundant information, which interferes with the classification results. Concurrently, as the value of c increases, the time required to train for 100 epochs also increases. Taking all factors into consideration, the model performs best at $c=3$, hence c is set to 3 in this work.

Note: To facilitate better observation and comparison of the data, the values of FDR and Time in this figure have been amplified by a factor of 100 and 10, respectively. Sensitivity-1 refers to the segment-based evaluation results, Sensitivity-2 denotes the event-based evaluation outcomes and Time refers to the time taken to train the model for 100 epochs.

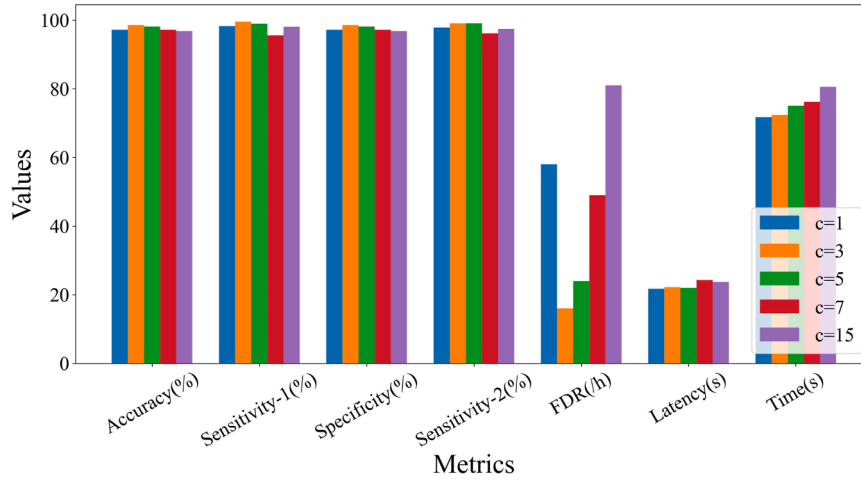


Fig. 11. Comparison of the metrics for the value of hyperparameter c.

Table 8

Complexity of various parts of the CNN-Informer model.

Sub-module	Parameters	MAC (Bytes)
CNN Layer 1	126	2.65 M
CNN Layer 2	126	653.19 K
CNN Layer 3	3.65 K	7.36 M
Informer Encoder 1	95.49 K	192.51 M
Informer Encoder 2	95.49 K	97.78 M
Informer Encoder 3 *	83.01 K	42.43 M
FC	590.72 K	18.90 M
Total	872.25 K	362.31 M

Note: * Without self-attention distilling operation.

5.3. Complexity analysis

Table 8 reports the complexity and memory usage of each part of the proposed CNN-Informer model as illustrated in Fig. 3. The parameter scale and memory access cost (MAC) of the CNN-Informer is 872.25 K and 362.31 M, respectively. The total parameters of the three-layer CNN amount to 3.90 K, with MAC of 10.66 M. Notably, compared to the Informer Encoder 3, the Informer Encoder 1 and the Informer Encoder 2 utilize self-attention distilling operations, each increasing the parameter number by 12.48 K. However, due to the halving of the self-attention matrix length, the overall MAC of the Informer model is 332.72 M and reduced by 33.62% compared to the version without distilling operations (501.27 M), significantly lightening the model's complexity.

The complexity comparison of the proposed CNN-Informer and CNN-Transformer is presented in Table 9. It can be found that the CNN-Transformer has 2.39 M parameters, while the CNN-Informer has only 0.87 M, reducing the number of parameters by 63.60%. In terms of computational complexity (measured in MAC), the CNN-Transformer consumes 580.23 M MAC, whereas the CNN-Informer requires only 362.31 M MAC, with a reduction of 37.56%. Overall, the time complexity of CNN-Informer decreases from $\mathcal{O}(L^2)$ to $\mathcal{O}(L \log L)$, L is the length of the input signal, and the model scale has been significantly reduced.

Table 9

Complexity comparison of different models.

Module	Parameters	MAC (Bytes)	Time Complexity
CNN-Transformer	2.39 M	580.23 M	$\mathcal{O}(L^2)$
CNN-Informer	0.87 M (63.60%↓)	362.31 M (37.56%↓)	$\mathcal{O}(L \log L)$

5.4. Compared with existing methods

Table 10 presents the results of several state-of-the-art seizure detection methods on the CHB-MIT scalp EEG dataset. In order to fully consider the spatial relationships between EEG channels, Zhao et al. (2021) utilized the Pearson correlation matrix of raw EEG signals as the input of a linear graph convolution network (LGCN) and used focal loss function to address data imbalance issues, achieving excellent performance in segment-based classification of ictal EEGs. However, their method required hand-crafted feature selection and lacked the event-based assessment, while our method achieves higher sensitivity in both segment-based and event-based evaluation without needing feature engineering. Wong et al. (2023) combined Out-Of-Distribution (OOD) and uncertainty estimation to detect seizures, achieving a segment-based specificity of 89.00%. Compared to our method, it exhibited lower performance and excluded event-based evaluation. Wang et al. (2024) used a fifth-order state-space dynamic system (SSDS) for feature extraction from the raw signals and then inputted these features into a decision tree (DT) classifier. Their method yielded accuracy of 94.1% and specificity of 97.5%, whereas their sensitivity was only 87.60%, far inferior to that of our model. Saranya and Bharathi (2024) decomposed raw EEG signals using Continuous Wavelet Transform (CWT) to extract manual features such as kurtosis, skewness, complexity feature, mobility feature, activity feature, entropy, weight feature, and bias. These liner and non-liner features were then fed into an improved adaptive neuro-fuzzy inference system along with a Light Gradient Boosting Machine (LightGBM) for seizure detection. This approach required time-consuming feature-engineering, and resulted in a more significant number of false positives and negatives. In addition, they did not conduct the event-based evaluation. In contrast, our method achieved efficient end-to-end seizure detection with segment-based sensitivity of 99.54% and event-based sensitivity of 99.07%. Tang et al. (2024) captured the dynamic dependencies relationships between different EEG channels using the Path Signature (PS) as distinguishing features, which were then fed into Bi-LSTM with attention mechanism for seizure detection, yielding a segment-based accuracy of 99.09%.

It is worth noting that all the studies mentioned above only conducted segment-based performance assessment, whereas event-based assessment is more clinically relevant. Li et al. (2020) employed a fully convolutional network with three convolutional blocks to extract robust features of seizure EEG data, which were then fed into Nested LSTM (NLSTM) to effectively learn hidden seizure characteristics and explore inherent temporal dependencies in EEG signals. Their method obtained a FDR of 0.66/h, much higher than our 0.16/h. Wang et al. (2021) combined random selection and data augmentation (RS-DA)

Table 10

Comparison of the performance for different methods on CHB-MIT dataset.

Author	Method	Segment-based			Event-based			Testing time for 1 h EEG (s)
		Accuracy (%)	Sensitivity (%)	Specificity (%)	Sensitivity (%)	FDR (/h)	Latency (s)	
Zhao et al. (2021)	LGCN+Focal Loss	99.30	98.82	99.43	—	—	—	—
Wong et al. (2023)	OOD	87.00	75.00	89.00	—	—	—	—
Wang et al. (2024)	SSDS+DT	94.10	87.60	97.50	—	—	—	—
Saranya and Bharathi (2024)	CWT+LightGBM	99.74	98.26	98.53	—	—	—	—
Tang et al. (2024)	PS+Bi-LSTM	99.09	99.28	98.95	—	—	—	—
	+Attention							
Li et al. (2020)	FC-NLSTM	95.29	95.42	95.29	94.07	0.66	—	—
Wang et al. (2021)	RS-DA+	99.54	88.14	99.62	99.31	0.20	8.10	—
	S-1D-CNN							
Zhang et al. (2022)	Bi-GRU	98.49	93.89	98.49	95.49	0.31	—	41.80
Zhou et al. (2023b)	STFT+CNN-LSTM	98.79	—	—	97.00	0.22	3.40	—
Dong et al. (2024b)	TCN-Bi-LSTM	97.09	94.31	97.13	96.48	0.38	—	5.65
This work	CNN-Informer	98.54	99.54	98.55	99.07	0.16	22.21	1.76

strategy to address sample imbalance and used a Stacked One-dimensional Convolutional Neural Network (S-1D-CNN) for seizure detection. Although this method attained excellent overall results, its segment-based sensitivity was much lower than ours. Zhang et al. (2022) used wavelet-transform to obtain specific frequency band relative energy of EEG signals as features and employed Bi-GRU for EEG classification. For event-based evaluation, the testing time was 42.80 s with an FDR of 0.31/h. Compared to this method, our method directly uses raw EEGs as inputs without the need for handcrafted feature extraction and performs better across all metrics. Zhou et al. (2023b) extracted the time-frequency information of EEG signals using STFT and constructed a feature matrix based on the spatial relationships of multiple channels. They employed CNN-LSTM as classifiers and achieved a latency of 3.40s. While this method demonstrated good real-time performance, its sensitivity and specificity in segment-based evaluation were not reported. Dong et al. (2024b) utilized the parallelism of Temporal Convolutional Network (TCN) and the long-range dependency mining capability of Bi-LSTM for automatic seizure detection. Although with the strength of its simple model structure, their model delivered longer testing time and higher FDR in comparison to our model.

As shown in Table 10, our method demonstrates superior performance across various performance metrics compared to other baseline models, indicating its potential for real-time application on seizure detection.

6. Conclusion and future work

In this work, a novel end-to-end model named CNN-Informer is presented for seizure detection in long-term EEG. The CNN-Informer model leverages convolutional module to extract EEG local features from multi-channel EEGs, and the ProbSparse self-attention mechanism of Informer to reduce the time and space complexity to the scale of $\mathcal{O}(\text{LlogL})$. Additionally, self-attention distilling operation further reduces the model size to mitigate overfitting caused by layer stacking. This combination of CNN and Informer allows the model to remain lightweight while focusing on key EEG signal features, thereby enhancing seizure detection performance. To validate the generalization ability of the proposed method, extensive evaluations are conducted on two scalp EEG datasets containing children and adults with epilepsy, respectively. On the CHB-MIT dataset, the model accomplished an accuracy of 98.54%, a sensitivity of 99.54%, a specificity of 98.55% in segment-based evaluation, and a sensitivity of 99.07% along with an FDR of 0.16/h and a latency of 22.21 s in event-based evaluation. Meanwhile, on the SH-SDU dataset, an accuracy of 92.78%, sensitivity of 94.40%, specificity of 92.92% in segment-based evaluation, and a sensitivity of 94.83% along with an FDR of 1.24/h and a latency of 33.81

s in event-based evaluation are achieved. These outstanding results demonstrate that the proposed CNN-Informer can effectively mine the local and long-range coupling of multi-channel EEGs for efficient seizure detection, and its lightweight architecture makes it suitable for real-time implementation.

In this study, the proposed seizure detection method is patient-specific. However, data variability between patients tends to be a significant challenge for cross-patient seizure detection, which will be the direction of our future research. Besides, most deep learning models have inherent “black-box” nature, limiting their interpretability and robustness, which may result in undesirable outcomes (Li et al., 2023b). Therefore, enhancement of the interpretability of our model will be our another future direction.

CRedit authorship contribution statement

Chuanyu Li: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haotian Li:** Writing – review & editing, Validation, Resources. **Xingchen Dong:** Writing – review & editing, Validation, Resources. **Xiangwen Zhong:** Writing – review & editing, Validation, Resources. **Haozhou Cui:** Writing – review & editing, Validation, Resources. **Dezan Ji:** Writing – review & editing, Validation, Resources. **Landi He:** Writing – review & editing, Validation, Resources. **Guoyang Liu:** Writing – review & editing, Validation, Resources. **Weidong Zhou:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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